

Elements of IoT

Hardware in IoT

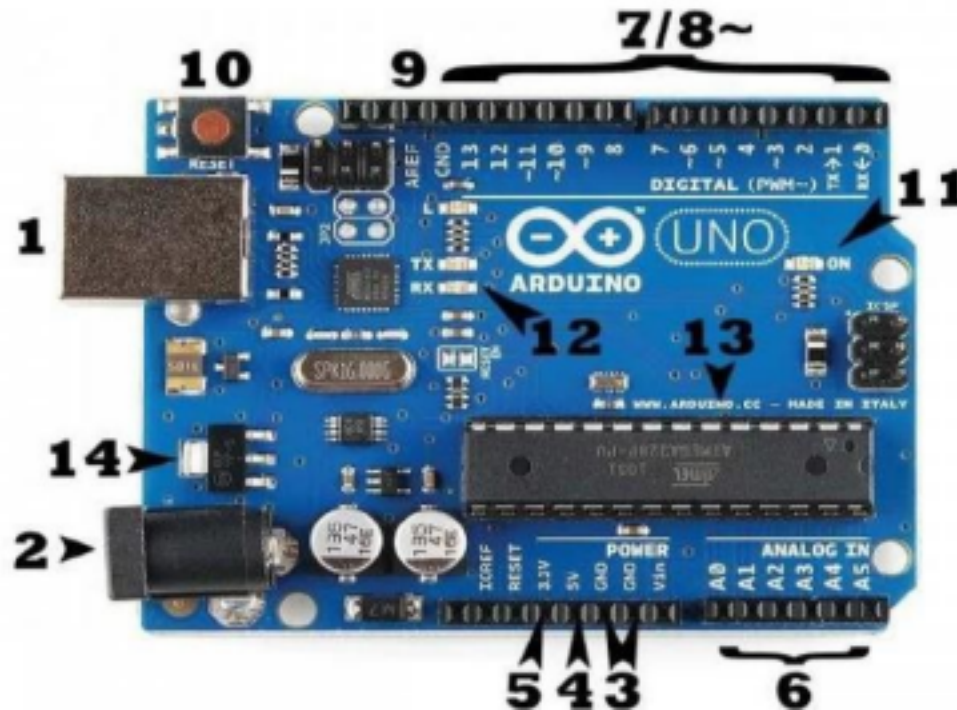
Arduino

- Arduino is an open-source platform used for building electronics projects.
- Arduino consists of both a physical programmable circuit board (often referred to as a microcontroller) and a piece of software, or IDE that runs on your computer,
- used to write and upload computer code to the physical board.

Hardware in IoT

Arduino

There are many varieties of Arduino boards that can be used for different purposes.



Hardware in IoT

Arduino

- **Power (USB / Barrel Jack) (2):** Every Arduino board needs a way to be connected to a power source.
- **Pins (5V, 3.3V, GND, Analog, Digital, PWM, AREF)**
 - ✓ **GND (3):** Short for 'Ground'. There are several GND pins on the Arduino, any of which can be used to ground your circuit. ✓
 - 5V (4) & 3.3V (5):** 5V pin supplies 5 volts of power, and the 3.3V pin supplies 3.3 volts of power. Most of the simple components used with the Arduino run happily off of 5 or 3.3 volts.
 - ✓ **Analog (6):** The area of pins under the 'Analog In' label (A0 through A5 on the UNO) is Analog In pins. These pins can read the signal from an analog sensor (like a temperature sensor) and convert it into a digital value that we can read.
 - ✓ **Digital (7):** Across from the analog pins are the digital pins (0 through 13 on the UNO). These pins can be used for both digital input (like telling if a button is pushed) and digital output (like powering an LED).

Hardware in IoT

Arduino

- ***Pins (5V, 3.3V, GND, Analog, Digital, PWM, AREF) ✓***

PWM (8): You may have noticed the tilde (~) next to some of the digital pins (3, 5, 6, 9, 10, and 11 on the UNO).

These pins act as normal digital pins, but can also be used for something called Pulse-Width Modulation (PWM).

✓ **AREF (9):** Stands for Analog Reference. Most of the time you can leave this pin alone. It is sometimes used to set an external reference voltage (between 0 and 5 Volts) as the upper limit for the analog input pins.

Hardware in IoT

Arduino

- **Reset Button(10):** Pushing it will temporarily connect the reset pin to ground and restart any code that is loaded on the Arduino. This can be very useful if your code doesn't repeat, but you want to test it multiple times.
- **Power LED Indicator:** Just beneath and to the right of the word "UNO" on your circuit board, there's a tiny LED next to the word 'ON' (11). This LED should light up whenever you plug your Arduino into a power source.
- **TX RX LEDs:** TX is short for transmit, RX is short for receive. These markings appear quite a bit in electronics to indicate the pins responsible for serial communication. In our case, there are two places on the Arduino UNO where TX and RX appear -- once by digital pins 0 and 1, and a second time next to the TX and RX indicator LEDs (12). These LEDs will give us some nice visual indications whenever our Arduino is receiving or transmitting data.

Hardware in IoT

Arduino

- **Main IC:** The black thing with all the metal legs is an IC, or Integrated Circuit **(13)**. The main IC on the Arduino is slightly different from board type to board type, but is usually from the ATmega line of IC's from the ATMEL company. This can be important, as you may need to know the IC type (along with your board type) before loading up a new program from the Arduino software. This information can usually be found in writing on the top side of the IC.
- **Voltage Regulator:** The voltage regulator **(14)** is not actually something you can (or should) interact with on the Arduino. But it is potentially useful to know that it is there and what it's for. The voltage regulator does exactly what it says -- it controls the amount of voltage that is let into the Arduino board.

Hardware in IoT

Raspberry pi

- The Raspberry pi is a single computer board with credit card size, that can be used for many tasks that your computer does, like games, word processing, spreadsheets and also to play HD video.
- It was established by the Raspberry pi foundation from the UK. □ It has been ready for public consumption since 2012 with the idea of making a low-cost educational microcomputer for students and children.
- The main purpose of designing the raspberry pi board is, to encourage learning, experimentation and innovation for school level students.
- The raspberry pi board is a portable and low cost. □ In the 20th century, the growth of mobile computing technologies is very high, a huge segment of this being driven by the mobile industries. The 98% of the mobile phones were using ARM

technology.

Hardware in IoT

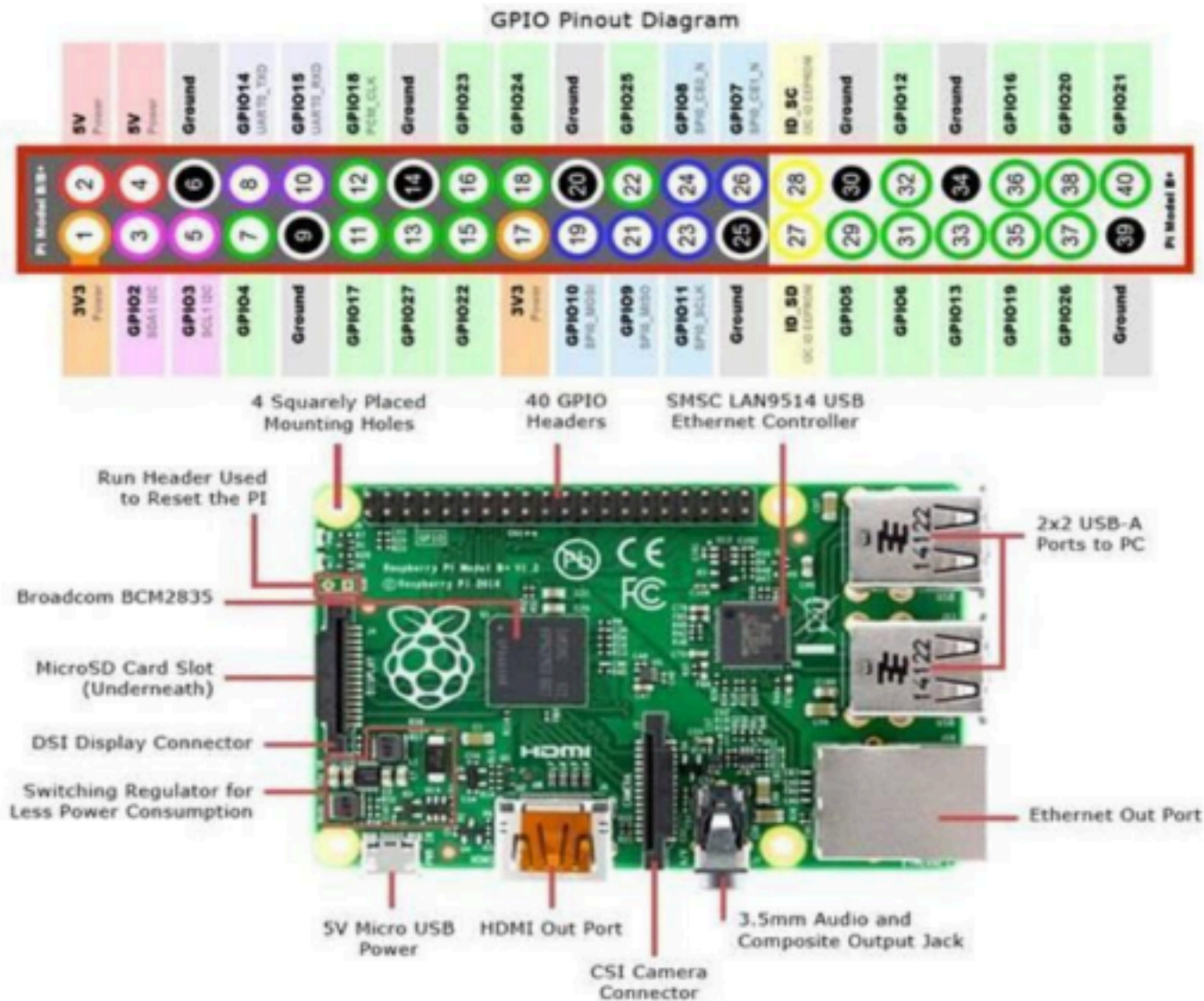
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Hardware in IoT

Raspberry pi



Hardware in IoT

Raspberry Pi Hardware Specifications

- The raspberry pi board comprises a program memory (RAM), processor and graphics chip, CPU, GPU, Ethernet port, GPIO pins, Xbee socket, UART, power source connector and various interfaces for other external devices.
- It also requires mass storage, for that we use an SD flash memory card. So that raspberry pi board will boot from this SD card similarly as a PC boots up into windows from its hard disk
- Essential hardware specifications of raspberry pi board mainly include SD card containing Linux OS, US keyboard, monitor, power supply and video cable.
- Optional hardware specifications include USB mouse, powered USB hub, case, internet connection, the Model A or B: USB WiFi adaptor is used and internet connection to Model B is LAN cable.

Hardware in IoT

- **Memory:** The raspberry pi model Aboard is designed with 256MB of SDRAM and model B is designed with 512MB. Raspberry pi is a small size PC compare with other PCs. The normal PCs RAM memory is available in gigabytes. But in raspberry pi board, the RAM memory is available more than 256MB or 512MB
- **CPU (Central Processing Unit):** The Central processing unit is the brain of the raspberry pi board and that is responsible for carrying out the instructions of the computer through logical and mathematical operations. The raspberry pi uses ARM11 series processor, which has joined the ranks of the Samsung galaxy phone.
- **GPU (Graphics Processing Unit):** The GPU is a specialized chip in the raspberry pi board and that is designed to speed up the

operation of image calculations. This board designed with a Broadcom video core IV and it supports OpenGL

Hardware in IoT

- **Ethernet Port:** The Ethernet port of the raspberry pi is the main gateway for communicating with additional devices. The raspberry pi Ethernet port is used to plug your home router to access the internet.
- **GPIO Pins:** The general purpose input & output pins are used in the raspberry pi to associate with the other electronic boards. These pins can accept input & output commands based on programming raspberry pi. The raspberry pi affords digital GPIO pins. These pins are used to connect other electronic components. For example, you can connect it to the temperature sensor to transmit digital data.
- **XBee Socket:** The XBee socket is used in raspberry pi board for the wireless communication purpose.
- **Power Source Connector:** The power source cable is a small

switch, which is placed on side of the shield. The main purpose of the power source connector is to enable an external power source.

- ***UART:*** The Universal Asynchronous Receiver/ Transmitter is a serial input & output port. That can be used to transfer the serial data in the form of text and it is useful for converting the debugging code.

Hardware in IoT

- ***Display:*** The connection options of the raspberry pi board are two types such as HDMI and Composite.
 - Many LCD and HD TV monitors can be attached using an HDMI male cable and with a low-cost adaptor.
 - The versions of HDMI are 1.3 and 1.4 are supported and 1.4 version cable is recommended.

- The O/Ps of the Raspberry Pi audio and video through HDMI, but does not support HDMI I/p.
- Older TVs can be connected using composite video. When using a composite video connection, audio is available from the 3.5mm jack socket and can be sent to your TV.
- To send audio to your TV, you need a cable which adjusts from 3.5mm to double RCA connectors.

Communication in IoT

- IoT is connection of devices over internet, where these smart devices communicate with each other, exchange data, perform some tasks without any human involvement.
- These devices are embedded with electronics, software, network and sensors which help in communication.

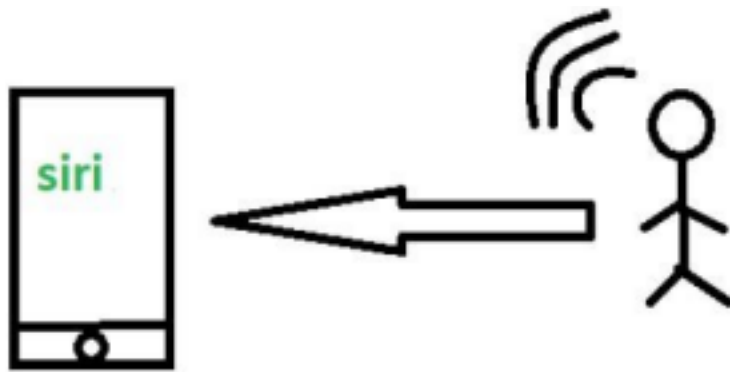
- Communication between smart devices is very important in IOT as it enables these devices to gather, exchange data which contribute in success of that IOT product/project.

Communication in IoT

Types of Communications in IOT:

a. Human to Machine (H2M):

- In this human gives input to IoT device i.e as speech/text/image etc. IoT device (Machine)like sensors and actuators then understands input, analyses it and responds back to human by means of text or Visual Display. This is very useful as these machines assist humans in every everyday tasks. It is a combo of software and hardware that includes human interaction with a machine to perform a task.



H2M communication

Communication in IoT

Types of Communications in IOT:

a. Human to Machine (H2M):

Merits: This H2M has a user-friendly interface that can be quickly accessed by following the instructions. It responds more quickly to any fault or failure. Its features and functions can be customized.

Examples:

- Facial recognition.
- Bio-metric Attendance system.
- Speech or voice recognition.

Communication in IoT

b. Machine to Machine (M2M):

- In this the interaction or communication takes place between machines by automating data/programs.
- In this machine level instructions are required for communication. Here communication takes place without human interaction.
- The machines may be either connected through wires or by wireless connection.
- An M2M connection is a point-to-point connection

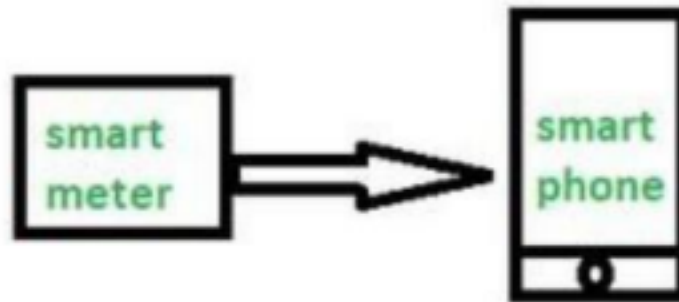
between two network devices that helps in transmitting information using public networking technologies like Ethernet and cellular networks.

- IoT uses the basic concepts of M2M and expands by creating large “cloud” networks of devices that communicate with one another through cloud networking platforms.

Communication in IoT

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M2M communication

Communication in IoT

b. Machine to Machine (M2M):

Merits:

- This M2M can operate over cellular networks and is simple to manage.
- It can be used both indoors and outdoors and aids in the communication of smart objects without the need for human interaction.
- The M2M contact facility is used to address security and privacy problems in IoT networks. Large-scale data collection, processing, and security are all feasible.

Demerits:

- ❑ In M2M, use of cloud computing restricts versatility and creativity. Data security and ownership are major concerns here.
- ❑ The challenge of achieving interoperability between cloud/M2M IoT systems is daunting.
- ❑ M2M connectivity necessitates the existence of a reliable internet connection.

Examples:

- Smart Washing machine sends alerts to the owners' smart devices after completion of washing or drying of clothes.
- Smart meters tracks amount of energy used in household or in companies and automatically alert the owner.

Communication in IoT

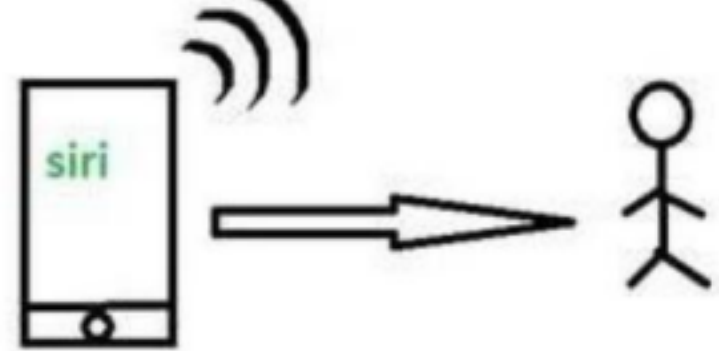
c. Machine to Human (M2H) :

- ❑ In this machine interacts with Humans. Machine triggers information (text messages/images/voice/signals) respective / irrespective of any human presence.
- ❑ This type of communication is most commonly used where machines guide humans in their daily life.

- ❑ It is way of interaction in which humans co-work with smart systems and other machines by using tools or devices to finish a task.

Examples:

- Fire Alarms
- Traffic Light
- Fitness bands
- Health monitoring devices



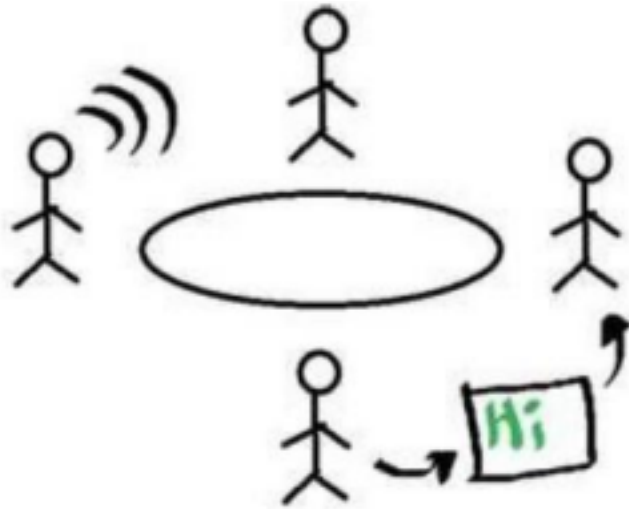
M2H communication

Communication in IoT

d. Human to Human (H2H) :

- ❑ This is generally how humans communicate with each other to exchange information by speech, writing, drawing, facial expressions, body language etc.

- ❑ Without H2H, M2M applications cannot produce the expected benefits unless humans can immediately fix issues, solve challenges, and manage scenarios.



H2H communication

Communication in IoT

❑ IoT I/O Interface

The process of connecting devices together so that they can exchange the information is called interfacing. In order for these devices to swap their information, they must share a

common communication protocol.

Generally, communication protocol can be separated two categories: parallel or serial.

❑ **Interfaces**

- ❑ **Parallel:** A parallel interface refers to a multi-line channel with each line capable of transmitting several bits of data simultaneously. They usually require **buses** of data — transmitting across eight, sixteen, or more wires. Data is transferred in huge, crashing waves of 1's and 0's.
- ❑ **Serial:** Serial interfaces stream their data, one single bit at a time. These interfaces can operate on as little as one wire, usually never more than four.

Communication in IoT

❑ **IoT I/O Interface**

- ❑ Serial interfaces have certain advantages over parallel

interfaces. ✓ The most significant advantage is simpler wiring. In addition, serial interface cables can be longer than parallel interface cables, because there is much less interaction (crosstalk) among the conductors in the cable.

- Most hardware interfaces are serial interfaces sacrificing potential speed in parallel. Serial interfaces generally use multiple wires to control the flow and timing of binary information along the primary data wire. Each type of hardware interface defines a method of communicating between a peripheral and the central processor.

Communication in IoT

□ IoT I/O Interface

IoT hardware platforms use a number of common interfaces. Sensor and actuator modules can support one or more of these interfaces:

- **USB.** Universal Serial Bus is a technology that allows a

person to connect an electronic device to a microcontroller. It is a fast **serial bus**.

- **GPIO**. General-purpose input/output pins area generic **pin** on an integrated circuit or computer board whose behavior including whether it is an **input** or **output pin** is controllable by the user at run time.
 - **GPIO pins** have no predefined **purpose**, and go unused by default. GPIO pins can be designed to carry digital or analog signals, and digital pins have only two states: HIGH or LOW.

Communication in IoT

□ IoT I/O Interface

- **I2C**. Inter-Integrated Circuit serial bus uses a protocol that enables multiple modules to be assigned a discrete address on the bus. I2C is sometimes pronounced “I two C”, “I-I- C”, or “I squared C”. I has two wires, a clock and data wire.

- **SPI.** Serial Peripheral Interface/Interchange Bus devices employ a master-slave architecture, with a single master and full-duplex communication.
- **UART.** Universal Asynchronous Receiver/Transmitter devices translate data between serial and parallel forms at the point where the data is acted on by the processor. UART is required when serial data must be laid out in memory in a parallel fashion.
- **RS 232** Recommended Standard 232 is used for obtaining communication between the computer and circuit such to transfer data between circuit and computer.

Protocols in IoT

MQTT protocol

□ MQTT stands for **Message Queuing Telemetry Transport**. MQTT is a machine to machine internet of things connectivity protocol. □ It is an extremely lightweight and publish-subscribe

messaging transport protocol.

- This protocol is useful for the connection with the remote location where the bandwidth is a premium. These characteristics make it useful in various situations, including constant environment such as for communication machine to machine and internet of things contexts.
- It is a publish and subscribe system where we can publish and receive the messages as a client.
- It makes it easy for communication between multiple devices. □ It is a simple messaging protocol designed for the constrained devices and with low bandwidth, so it's a perfect solution for the internet of things applications.

Protocols in IoT

ZIGBEE

- ZigBee is a Personal Area Network task group with low rate task group 4. It is a technology of home networking. ZigBee is a technological standard created for controlling and

sensor the network. As we know that ZigBee is the Personal Area network of task group 4 so it is based on IEEE 802.15.4 and is created by Zigbee Alliance.

- ❑ ZigBee is a standard that addresses the need of very low-cost implementation of Low power devices with Low data rate for short-range wireless communications.

Types of ZigBee Devices:

- **Zigbee Coordinator Device** – It communicates with routers. This device is used for connecting the devices.
- **Zigbee Router** – It is used for passing the data between devices.
- **Zigbee End Device** – It is the device that is going to be controlled

Protocols in IoT

Bluetooth

- ❑ Bluetooth has been in the tech market as a wireless channel of connection between devices since Ericsson invented it in 1994.

Since then, Bluetooth technology has evolved and has become the go-to wireless connectivity solution for wearable's, gadgets, and other devices.

❑ Nowadays, we can find Bluetooth everywhere; cars, speakers, wearables, medical devices, wireless headphones, shoes, etc. ❑ In any modern device, it is safe to assume that you have encountered and used Bluetooth technology at one point or the other. ❑

Bluetooth is a short-range wireless technology medium used for exchanging data between two electronic devices (usually mobile) over a short distance.

- ❑ This process completely eliminates the primitive use of cables for connectivity.
- ❑ A typical example is how you can listen to music with a headset on the go without having to plug it into the headset jack of your mobile device.

Protocols in IoT

Constrained Application Protocol (CoAP)

- ❑ CoAP a customary client-server IoT protocol. It enables clients to

make requests for web transfers as per the need of the hour. On the other hand, it also let supporting servers to respond to arriving requests. In summary, devices' nodes in the IoT ecosystem are enabled to interact over through CoAP only.

- ❑ CoAP and HTTP follow the same working procedure. However, CoAP attains its functionality via asynchronous transactions (using UDP). It utilizes the POST, GET, PUT, and DELETE calls. That's the reason why API security is of higher grade while CoAP is active as it is an RPK and PSK certified protocol.

CoAP is compatible with 4 types of information exchange:

1. Acknowledgments confirm the completion or failure of an event.
2. Confirmable are the messages that are resent on time out until the confirmation of successful sending doesn't arrive.
3. Reset messages are empty, with confirmable as their nature.
4. Non-confirmable information is just sent and has no guarantee of successful delivery. There is no acknowledgment of success either.

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Protocols in IoT

TCP/IP and UDP

- ❑ The TCP/IP protocol despite being most common protocol stack on internet is not much suitable for IoT applications due to large overhead.
 - ❑ It is more suitable for applications where reliable delivery of data with high bandwidth in hand is required.

- The IoT applications generally have limited bandwidth and need swift transfer of small data packets.
- In such case, the UDP/IP stack is far better than TCP/IP.
- The User Datagram Protocol (UDP) is the simplest transportation layer protocol used primarily for establishing low-latency and loss tolerating connections between applications on the communication network.
- Both TCP and UDP run on the top of Internet Protocol (IP) that is why they are referred as TCP/IP and UDP/IP.

Protocols in IoT

- UDP is a connectionless protocol which means the sender just transmits the data without waiting for the connection with the receiver.
- It is an unreliable protocol when compared with TCP.
- There is no error checking mechanism or correcting mechanism involved in data transmission which results in

using less bandwidth.

- UDP protocol just sends the packets (or datagram).

- There is no acknowledgement guarantee of packet received by the other end.

- It allows for less data overhead and delays.